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ORIGINAL RESEARCH

Role of Artificial Intelligence Capability in the Interrelation Between Manufacturing Strategies and Operational Resilience

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Abstract *Recent global disruptive events, like COVID-19, energy transition, war, and recession, affect the firm's operations. Manufacturing firms recently focused on resilience and moved toward adopting new technologies. This study identifies the role of artificial intelligence capability (AIC) between manufacturing strategies, namely production strategy (PRS), product development strategy (PDS), and service quality strategy (SQS), and firms' competitive advantage in terms of operational resilience. A resource-based view has been utilized to develop AIC in manufacturing firms for operational resilience, and the hypotheses were examined using the structural equation modeling. Two hundred thirteen respondents participated in the survey, including five constructs and twenty-nine items. The findings reveal that product development strategy is linked beneficially to production, service quality, and AIC. The production strategy is also linked to the SQS. The findings also confirm that AIC mediates between PDS, PRS, and SQS and operational resilience. The study is significant for practitioners in understanding the role of AIC in modernizing manufacturing and production strategies for firms' competitive advantage as firms become more operationally resilient. This study is unique because it empirically analyzes the role of manufacturing strategies in building AIC for improving operational resilience as a competitive advantage for a firm.*

Keywords Artificial intelligence capability · Manufacturing strategies · Operational resilience · Structural equation modeling

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Introduction

Recent developments and major global events like COVID-19, cyber-attacks, and climate and biodiversity concerns owing to misuse of natural resources have generated concern about supply chain resilience. These events and crises significantly impact present and future supply chain management (SCM) (Alvarez et al., 2018; Crippa et al., 2021; Sharma et al., 2023). The COVID-19 pandemic, Brexit, and recent computer shortage have identified global supply chain weaknesses due to uncertainties in logistics lead times, labor availability, product regulations, trade tariffs, and various other fields. Therefore, highly disruptive events can suddenly accelerate demand or create stockout products (Kim et al., 2021), threatening continuous production, flow, and fulfillment of consumer demand. There are several instances of how the COVID-19 outbreak has caused difficulties in industrial companies (Hasan et al., 2023; Mokline & Ben Abdallah, 2022; Wided, 2023). For example, Kojima Press Industries, a local supplier to Toyota, had to close its servers due to COVID-19-related problems with internal systems. Consequently, Toyota's ordering system malfunctioned, resulting in the total closure of all Toyota factories in Japan (Toyota, 2022). In February 2020, Hyundai also ran out of Chinese components, forcing it to close its South Korean production facilities (White et al., 2020). The lack of raw resources has caused the global industrial sector's production output to drop significantly to 11.2 percent. During the epidemic, the lack of raw materials, labor shortages, fluctuating demand, transportation stoppages, and government controls impeded the operation of industrial facilities (Dohale et al., 2023).

Operations are the prime process of industries in providing products as per demand. Thus, production operations must be resilient to disruptions to continue a sufficient supply of goods (Cheng et al., 2022). Businesses must reorganize their production systems, SC procedures, and operations to address the detrimental effects of disruptions like COVID-19. This emphasizes the need for increased resilience, which business practitioners and researchers have highlighted as an effective way to overcome interruptions and decrease SC risk (Hendry et al., 2018; Queiroz et al., 2022; Roscoe et al., 2020). The capacity of a system to adjust to and handle changes brought about by unforeseen occurrences while preserving the system's fundamental functions and structure is known as resilience (Holling, 1973). Resilience has evolved as a significant aspect of supply chain (SC) vulnerability and risk management (Adobor & McMullen, 2018). It allows firms to maintain continuity in their operations and services to consumers (Ambulkar et al., 2015). Research interest is growing in the field of resilience, showing that resilient organizations have structures for handling disruptions to achieve higher performance (Wong et al., 2020). Therefore, there is a need for operations and production systems to depend upon innovative technologies and strategies to be more resilient and to survive (Kumar et al., 2022b).

Considering continuous and radical changes in the market and technology, higher innovation performance is required to make firms more resilient in the current business environment. Due to rising information asymmetry, firms must expand their boundaries beyond knowledge generation mechanisms to improve innovation performance (Chesbrough & Bogers, 2014). Research is required to determine how the integration of production systems affects innovation performance, which results from a product creation strategy. Innovation is influenced by total quality management (TQM), quality management, labor flexibility, and strategic flexibility (Miroshnychenko et al., 2021). Therefore, the interaction between resilience, production strategy, service quality, and product development strategy must be investigated. The evolution of manufacturing strategies slowed down after 2005 and has been limited to manufacturing strategy in relation to performance and competitiveness. It has lacked the interaction of resilience with manufacturing strategies in the context of artificial intelligence (AI) adoption in a manufacturing system.

Businesses must reorganize their production systems, SC procedures, and operations to address the detrimental effects of disruptions like COVID-19. This emphasizes the need for increased resilience, which business practitioners and researchers have highlighted as an effective way to overcome disruptions and reduce SC risk (Queiroz et al., 2022; Roscoe et al., 2020). The capacity of a system to adjust to and handle changes brought about by unforeseen occurrences while preserving the system's fundamental functions and structure is known as resilience (Holling, 1973).

While adopting AI depends on the firm's technological capability, resource requirement, and availability (Mohiuddin et al., 2022), based on the resource-based view (RBV), manufacturing strategies are the firm's assets and work as intangible resources. By connecting technology with SC resilience in manufacturing and production, we can rethink and improve resilience in today's global SC (Wieland & Durach, 2021). The COVID-19 crisis revealed that the connection of SC research with other fields has not been considered significant in the past (Wieland., 2021). The COVID-19 outbreak emphasized adopting smart technologies and implementing SC crisis mitigation strategies (Khan et al., 2022). The crisis also revealed the importance of preparedness in firms' operations. Therefore, it is time to excel in our knowledge of resilience and its relation to manufacturing strategy in technology applications, e.g., AI adoption. By providing answers to the following research questions (RQs), this study fills in the gap in the literature:

RQ1: How does product development strategy influence production and service quality strategies?

RQ2: How does production strategy influence service quality strategy?

RQ3: How do production, product development, and service quality strategies contribute to building AI capability in the manufacturing industry?

RQ4: How does AI capability enhance the competitiveness of manufacturing firms in terms of operational resilience?

This study's primary contribution is to present an RBV approach-based theoretically grounded research model for building AI capabilities in manufacturing industries for competitive gain through improvement in operational resilience. With the SEM methodology, we empirically prove how the firm's AI capability interrelates between manufacturing strategies (product development strategy, production strategy, and service quality strategy) and the firm's operational resilience. This study will provide direction to astounding managers interested in understanding how AIC can maximize the benefit of enhanced operational resilience through successful manufacturing strategies.

The manuscript is structured as follows. Section "Literature review" gives an in-depth literature background; while, Section "Theoretical background and hypotheses development" represents the development of hypotheses based on the preceding study's theoretical basis. In Section "Research methodology", the research methodology is discussed, and in Section "Data analysis and results", the research findings are analyzed in-depth. In Section "Discussion and implications", the findings are explored and the implications for theory and practitioners are stated. Finally, the study is concluded in Section "Conclusions, limitations, and future research directions".

Literature Review

Previous literature, as presented in Table 5 Appendix A, has discussed AI-driven SC analytics, operational performance (Dubey et al., 2021a), AI practices, service experience, customer engagement and non-economic advantage (Baabdullah et al., 2021), constructs of manufacturing strategy, integration (Qi et al., 2017), environmental factors, process factors, information sharing, AI and SC risk mitigation (Nayal et al., 2021), operations strategy/manufacturing strategy, SC integration, innovation performance (Kumar et al., 2020), SC resilience (Belhadi et al., 2021b), product development, and manufacturing process (Li et al., 2005). Some authors have explained role of AI in improving SC resilience (Modgil et al., 2022), the adoption of emerging technologies in improving SC resilience (Chatterjee et al., 2022), and the role of supply chain analytics in improving SC risk management strategies, namely agility, alignment, and adaptability (Khan et al., 2023), data analytic capability for improving SC resilience and sensitivity through improving expectation and innovation capability (Munir et al., 2022). How product development, manufacturing, and service quality methods affect operational resilience has received less attention. Moreover, there is a paucity of studies on how interaction among manufacturing strategies will affect AI's capability to improve operational resilience. There is also an absence of evidence and information concerning the integration of AI technologies, features, advantages, problems, and resilience in production and operations systems (Spanaki et al., 2022).

Product Development, Production, and Service Quality Strategies

A competitive advantage is the source of new product development (Brown et al., 1995). Based on prior research, PDS variables are categorized into structure and process. Market competitiveness, market growth and size potential, firm strategic characteristics, resources, and product development organizational structure are structure-based elements that influence or are involved in product development. Process-related factors include supporting factors such as top management support, consumer and supplier participation, information sharing involving cross-departmental communication, team formulation, and decision-making methods such as front-end loading and stage-gate approach (Schoenherr & Wagner, 2016). Moreover, the level of uncertainty is the more significant factor affecting product development in the current scenario. Without a highly qualified and skilled cross-functional team, a company fails to develop successful products if uncertainty exists (Morita & Machuca, 2018). Uncertainty is classified into four categories: market uncertainty, product uncertainty, process uncertainty, and resource uncertainty. These uncertainty sources can be novelty, multiplicity, validity, and reliability of information (Brun et al., 2009). Product development strength may also be determined by the integration of useful information and development activities. The firm's shared product perception is crucial for higher product development performance (Tessarolo et al., 2007). Product development is mainly studied in literature to align it with SC processes (Marsillac & Roh, 2014).

Production strategy refers to the cost and flexibility skills that a company develops. It is also one of the firm's business strategies (Anderson et al., 1989). Production strategy improves competitive advantage (Ward & Duray, 2000). When well-formed and implemented while considering internal and external factors, production strategy improves competitiveness by successfully deploying competitive production capabilities such as cost and flexibility (Swamidass, 1986). Turning manufacturing processes into a source of competitive edge is essential for a successful product strategy (Ehie & Muogboh, 2016). Production strategy as a variable with two dimensions, namely flexibility and cost, has not been studied in relation to product development and service quality strategy. There is, thus, a need for this to be explored rigorously. Operation strategy can enhance SC integration (Kumar et al., 2020). Product strategy combines information and managerial actions (Paiva et al., 2008). Firms with innovative products emphasize flexibility as an effective production strategy and fewer cost strategies (Miller & Roth, 1994).

Service quality strategy is a competitive requirement consisting of quality, delivery, and service performance parameters. Quality includes aspects of the performance of products, durability, and reliability of products. It depends on manufacturing capabilities, design blueprints, and the firm's capacity (Ehie & Muogboh, 2016). Product reliability measure is included in this paper as one of the quality measures. Delivery is considered a competitive parameter after the supply chain becomes significant in operations. Delivery performance is evaluated based on delivery time and volume. Moreover, firms adopting lean manufacturing, just-in-time, and zero inventory are considered competitive preferences. The delivery reliability for the delivery parameter in this study represents firms' capability to provide on-time delivery, adjust order sizes without impacting delivery performance, and align delivery schedules per each consumer's needs. The low frequency of consumer backorders also comes under the delivery parameter (Kaur et al., 2017). Service parameters facilitate product use and modification to consumers' demands, use, and behavior. Service involves consumer value co-creation and intense contact. These are relationship-based, complex, and personalized (Sousa & Da Silveria, 2017). A high service level increases profit because of its differentiated nature (Sousa & Da Silveria, 2017). Implementation of services includes managing consumer processes for fulfilling, demanding, and complex consumer needs (Sousa & Da Silveria, 2017). The relational approach of service can allow manufacturing firms to affect consumer processes, further improving adjustment between the manufacturer and consumer processes (Payne et al., 2008).

Resource-Based View (RBV)

The RBV theory has gained significant prominence as a theoretical background to describe the ongoing gaps in performance observed among firms involved in strategic research (Barney & Griffin, 1992). Based on the RBV theory, organizations possess a collection of unique, rare, non-substitutable, and difficult-to-replicate resources and capabilities, which can confer a sustained competitive edge over time. Consequently, resources pertain to both actual and intangible assets owned or controlled by the firm. On the other hand, capabilities refer to a company's capacity to make efficient use of and connections among its resources (Laosirihongthong et al., 2014). Physical capital resources, such as information technology (IT) infrastructure, workforce, and institutional capital resources, are the three types of resources. Analyzing how competitive strategy affects internal and external resources and their influence on an organization's capabilities is of utmost importance (Laosirihongthong et al., 2014).

Artificial Intelligence Capability

Artificial Intelligence

'Artificial intelligence' is a set of tools and techniques to help computers and other devices learn from experience and new information. AI also manages cognitive services that help humans perform complicated tasks (Grover et al., 2020; Kar et al., 2023). AI helps robots mimic human activities in industrial processes to model real-world scenarios (Belhadi et al., 2021a). A lot of data, finer algorithms, and better processing hardware impact the development of AI in several domains (Brynjolfsson & McAfee, 2017). Leading tech businesses are interested in AI tools (Belhadi et al., 2021a). AI mimics human cognition and performance. According to Chakravorti et al. (2019), this entity may learn and make its own judgments, which may help or replace human cognition in cognitive tasks. Wilson and Daugherty (2018) say AI can improve responsiveness, customization, scalability, innovation, flexibility, and decision-making. Wang et al. (2021) explored how AI affects product manufacturing and discussed AI-enabled material acquisition, production planning, resource configuration, machining, assembly, logistics, quality control, and storage.

Artificial Intelligence Capability

AI has garnered significant interest in augmenting the firms' competitive edge in industrial enterprises. According to RBV theory, a manufacturing firm's AI competency is its ability to create, integrate, and apply AI-based resources and strategies to gain a competitive edge (Chowdhury et al., 2022). However, AI technology enterprises must use resources to get a competitive edge. According to the RBV for building a firm's AI capability, firms have internal and external resources. The internal and external firm's resources are classified as tangible, intangible, and human skill resources. The tangible resources required to develop AI capability include financial support, data, technical support, software and hardware, and technological infrastructure (Chowdhury et al., 2022). In comparison, intangible resources have technical skills, human resources, knowledge, governance, and regulations.

Furthermore, using AI in manufacturing promotes significant advancements in fault prediction, intelligent decision-making, human–machine cooperation, and the identification of product design flaws. AI makes it possible to model production systems, make decisions, and integrate data throughout the manufacturing process. Thus, incorporating AI into production systems guarantees a significant advancement in intelligent manufacturing in the future (Wang et al., 2021).

Operational Resilience

Scholten et al. (2019) have identified disruption absorption, recoverability, adaptability, and transformability as the key dimensions of resilience at the output level when dealing with disruptions. The notion of disruption profile posits that by ascertaining the baseline operational performance level before an interruption, one may effectively gauge the resilience of operations in the face of a disturbance. The ways to quantify operational resilience are (1) determining how much the standard operational performance level fell immediately following a disruptive incident before recovery efforts start and (2) determining how long it takes the company to return operations to an ordinary performance level (DesJardine et al., 2019; Li et al., 2020). According to Blackhurst et al. (2011), a larger decline in ordinary performance level signifies that the operations are not capable of being disrupted; while, a smaller decline in usual operations suggests that the operations are capable of being disrupted. According to Li et al. (2020), a surgery is considered unrecoverable if its recovery period is prolonged. The capacity of a firm's operations to endure and bounce back from supply chain shocks is the official definition of operational resilience. The ability of an organization to preserve its operational framework and carry on with business as usual during disruptions is known as the disruption absorption aspect. According to Essuman et al. (2020), recoverability refers to an organization's capacity to resume operations in a pre-disturbance normal performance condition. A company's ability to absorb disruptions does not imply that it can recover from them (Behzadi et al., 2017). Strong operational resilience is necessary for any business for excellent resistance and recoverability, making it a significant competitive advantage (Kumar et al., 2022b; Sánchez-García et al., 2023).

Theoretical Background and Hypotheses Development

Theoretical Background

In the literature review, the RBV is utilized to develop AI capability through successful strategies such as PRS, PDS, and PQS to advance a firm's operational resilience. According to RBV, product development strategy and knowledge utilization from collected information can be considered unique intangible strategic resources. These interactions can increase competitive advantage and collateral performance (Barney, 1991). The interconnection of manufacturing strategy, information management technology, and operational resilience is crucial to business success, particularly during periods of uncertainty or disruptions (Veloso Saes et al., 2022). Based on an extensive analysis of past literature and RBV (Dubey et al., 2021b; Kumar et al., 2020; Li et al., 2005; Nayal et al., 2021; Qi et al., 2017), the conceptual framework has been proposed as presented in Fig. 1. Figure 1 reveals the role of manufacturing strategies driven by AI capability to gain operational resilience as a firm competitive advantage.

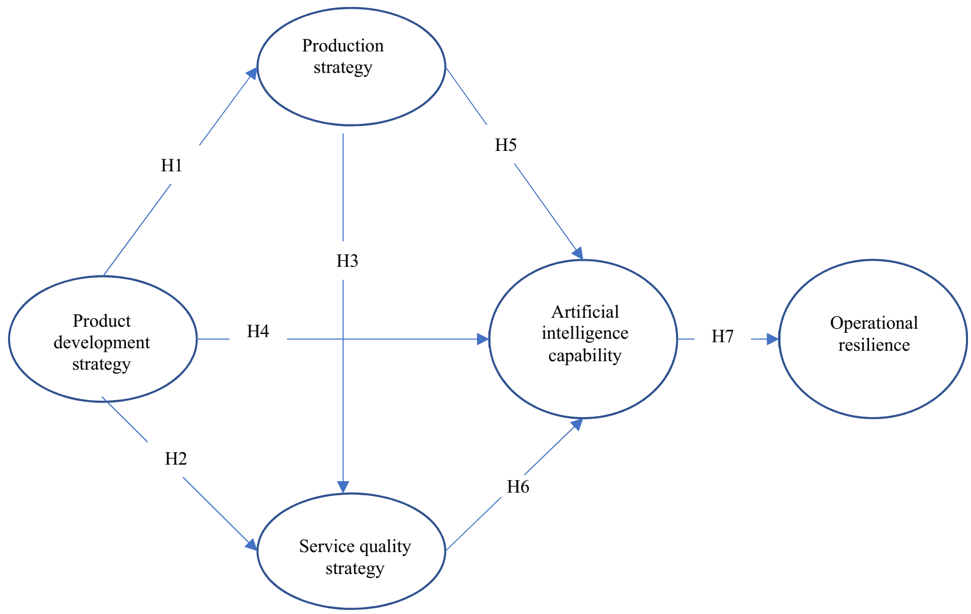
Product Development, Production, and Service Quality Strategies

Previously, the linkage between manufacturing and product development strategies has not been explored. Kumar et al. (2020) examined how operations strategy, including cost strategy and flexibility, affects innovation performance. Li et al. (2005) explored how product development affects production. Few studies explored the link between manufacturing and product development strategies. According to Legre-Vidal et al. (2004), operations strategy and innovation are linked. More innovative industries emphasize flexibility, whereas less innovative industries focus on delivery (Alegre-Vidal et al., 2004). Industries with innovative products prioritize flexibility over cost strategies (Miller & Roth., 1994). A product development strategy is applied to improve innovation performance. In this study, production strategy includes dimensions of flexibility and cost strategy. Production strategy needs to be aligned with product development capability by developing and implementing an effective product development strategy. Thus, companies need to adopt flexible and cost-effective production systems to launch new products faster with an effective product development strategy, thus reducing setup time and cost.

H1 Product development strategy is positively linked with a successful production strategy.

Kumar et al. (2020) discussed the effect of operations strategy having components of service quality strategy (e.g., quality and delivery) on innovation performance. Li et al. (2005) explored the relationship between product development and service quality components by investigating the effect of product development on the manufacturing process. Sousa et al.. (2020) investigated the effect of service differentiation advantage on product differentiation advantage. Designing and introducing new products can result in higher service quality. The manufacturing process, which also includes elements of service quality strategy, is a fundamental component for extracting maximum benefits from product development (Li et al., 2005). Service quality and delivery are two examples of operational methods that have been shown to affect a company's ability to innovate (Kumar et al., 2020). Alegre-Vidal et al. (2004) also showed a positive relationship between

Fig. 1 Conceptual model



operations strategy (delivery, quality) and innovation. More innovative firms focus on quality and less innovative firms focus on delivery (Alegre-Vidal et al., 2004). Quality improvement can enhance firms' innovation performance. To achieve a higher level of quality, organizations need to increase their innovation performance to successfully launch new products and services (Kumar et al., 2020).

H2 Product development strategy is positively linked with a successful service quality strategy.

Bortolotti et al. (2015) investigated the relationship of delivery performance with flexibility performance. Improvement in delivery performance reduces the time needed to respond to changes in consumer demand, thus supporting process alignments according to varying demands (Größler & Grübner, 2006). High delivery performance indicates strong relationships between firms and suppliers, leading to suppliers' ability to adapt to demand variations (Rosenzweig & Roth, 2004). Thus, flexibility is improved by reducing response time, building strong supplier relationships, and improving delivery performance. Improving quality makes production processes stable and removes reprocessing work, thus reducing energy and resource consumption (Rosenzweig & Roth, 2004) and overall cost. Reducing lead time reduces inventories and transportation time, thus improving delivery reliability (Koufteros et al., 2002). Avella et al. (2011) concluded that quality indirectly affects flexibility by improving delivery.

H3 Production strategy has a positive linkage with a successful service quality strategy

Manufacturing Strategies and Artificial Intelligence Capability

The available literature has not directly discussed the relationship between product development strategy and AIC. Usually, AI technologies and tools can improve innovation performance (Wilson & Daugherty, 2018). Smarter choices may be made using big data analytics (BDA) during product development. There has been less literature on how AI interacts with the various parts of PDS (Nayal et al., 2021). Organizational elements, such as management support and cooperation associated with PDS, were studied by Nayal et al. (2021) to determine their effect on AI. Information exchange and SC integration with AI applications were found to have a favorable correlation by Nayal et al. (2021). An association between innovative output and SC integration was found by Kumar et al. (2020). AI capability promotes competitiveness, integration, and innovation (Borges et al., 2021). Information sharing, integration, and competitiveness are components of PDS (Rahman et al., 2022). Suppose these components can have a significant relationship with AI capability. In that case, there is a need to empirically study the effect of PDS on AI capability by including extensive measures of PDS. In developing AI capability, product development strategy is a strategic data resource because product development has a large volume of data, and handling data requires AI skill knowledge resources.

H4 Successful product development strategy positively relates to firms' AI capability.

Qi et al. (2017) explored the indirect effect of flexibility and cost strategies on internal and external integration through lean and agile SC strategies. Kumar et al. (2020) examined the effect of operations strategy, including components of production strategy, on SC integration. Kumar et al. (2020) found that flexibility positively affects SC integration but that cost does not significantly affect SC integration. Nayal et al. (2021) explore the effect of process factors on AI applications. Process factors involve inventory management related to the flexibility component of production strategy. AI-enabled automation can help improve efficiency, thus reducing costs. Qi et al. (2017) found that cost and flexibility strategies indirectly affect SC integration. Thus, production cost and flexibility strategy components can impact AI capability that promotes the integration of suppliers and customers (Borges et al., 2021).

H5 Successful production strategy has a positive relationship with firm's AI capability.

Dubey et al. (2021a) investigated the impacts of AI-enhanced supply chain analytics (SCA) on operational performance, focusing on factors like on-time delivery and order fulfillment lead time. In order to contribute or create value based on the consumer's perspective, the service must be tailored to their specific wants, desires, and purchasing habits (Ameen et al., 2021). Kumar et al. (2020) observed that delivery does not significantly influence SC integration, while quality strongly affects SC integration. Process and environmental parameters were shown to considerably favorably influence AI applications (Nayal et al., 2021). These problems entail quality difficulties and delivery delays connected to service quality measures. The author discovered a strong correlation between process variables and defects in AI. If the correlation between delivery and AI turns out to be insignificant, it is worth double-checking the impact of external factors like delivery delay. Raut et al. (2019) found a non-significant relationship between quality management and BDA and AI components. Qi et al. (2017) found an indirect positive effect of delivery and quality strategy on SC integration, which can be enhanced by AI applications. Thus, the effect of service quality consists of both constructs, namely quality and delivery service. Wang et al. (2021) also highlighted the role of AI capability in the product service stage in terms of intelligent consumer service, customization, product status monitoring, and intelligent recycling.

H6 Successful service quality strategy has a positive relationship with firm's AI capability.

Artificial Intelligence Capability and Operational Resilience

The effect of AI capabilities on operational resilience has not been experimentally investigated in prior work. A few writers have highlighted the function of AI capabilities in managing resilience to reduce the risks associated with SC (Baryannis et al., 2019; Nayal et al., 2021). The ability of a business to recover from interruptions is known as operational resilience (Van Der Vegt et al., 2015). Returning to regular operations means that deliveries and sales have not been disrupted. Belhadi et al. (2021b) discovered a strong correlation between AI and SC resilience and collaboration. According to Modgil et al. (2021a), AI makes it possible for information processing skills to recognize, evaluate, reorganize, and activate in order to guarantee SC resilience in the face of disruptions like COVID-19. According to Modgil et al. (2021b), AI may support supply chain resilience by facilitating features including transparency, last-mile delivery, impact minimization, agile procurement, and customized solutions for all parties involved. Therefore, in order to comprehend AI's performance and risk-reduction function, we want to investigate how its competence affects operational resilience.

H7 AI capability of manufacturing firms is positively linked to firm's operational resilience.

Research Methodology

A conceptual framework is developed to validate the AI-driven supply chain operational resilience strategy. This study has been divided into four phases. The first is the development of a conceptual framework and the identification of factors and items. The second phase has sample characteristics, data collection, and validation. The third phase tests the reliability of obtained data and factor validation with items using exploratory factor analysis (EFA) in IBM SPSS v.20. The fourth phase tests hypothesis and model validation in AMOS v.20. The research framework followed in this study is shown in Fig. 2.

Extraction of Factors and Items

First, the authors comprehensively searched the literature on various scientific databases for operational resilience strategies. The literature review identified and selected five factors and 29 related items. The identified literature was discussed with experts and finalized. The five factors are production strategy, product development strategy, service quality strategy, AI, and operational resilience.

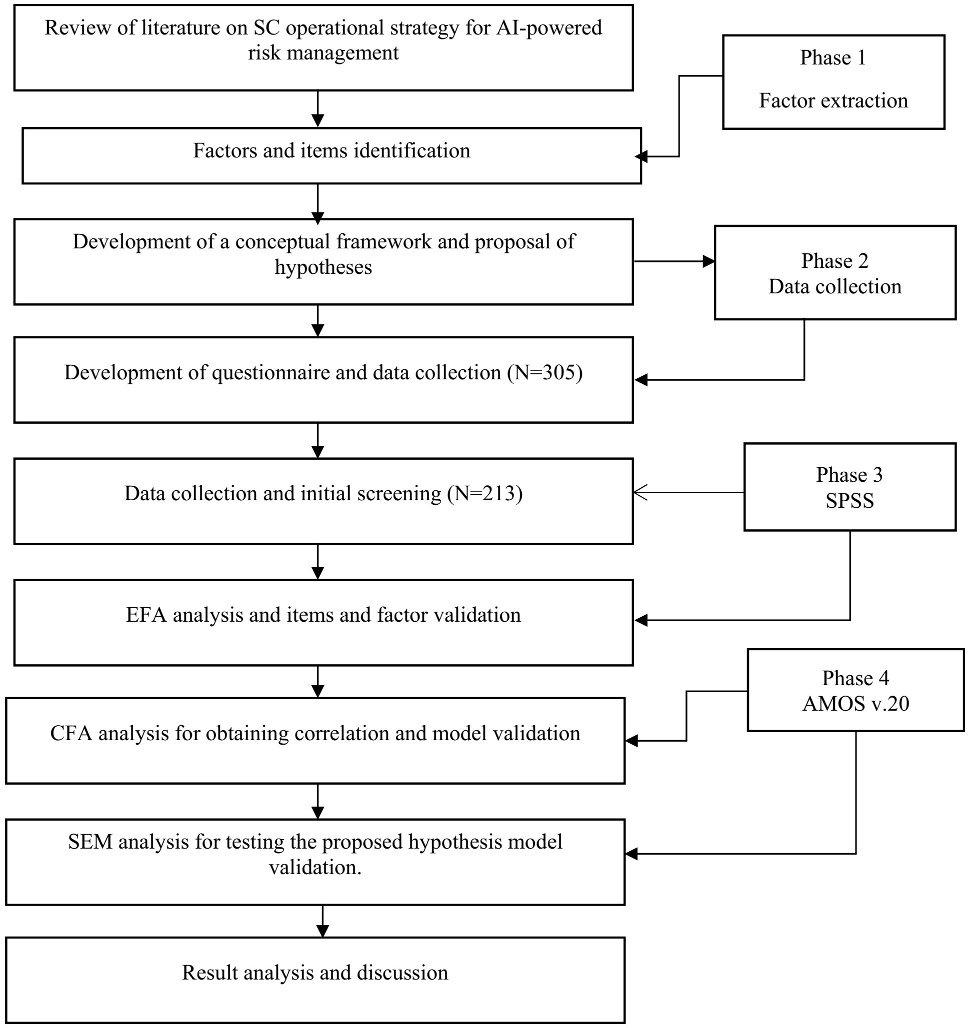


Fig. 2 Research framework

Questionnaire Development and Data Collection

A survey questionnaire that included 29 questions was framed. A 5-point Likert scale, with 1 representing “strongly disagree” and 5 representing “strongly agree,” was used to create the questionnaire. The questionnaire underwent preliminary testing before being distributed to ten industry experts from various firms. The participants were requested to participate in the survey while being observed by the researcher and were also invited to provide their recommendations for improving the survey. Table 5 of Appendix B contains the measuring items and the questionnaire. After that, we emailed the questionnaire to 420 responders from various industries and academic institutions. About 160 persons answered the survey within a month of its distribution. A second reminder is sent after a month for those who have not replied. Sixty additional replies came in after the follow-up mailing. Hence, a total of 220 responses, nearly 52.3 percent, were collected. From 220 obtained responses, 213 were satisfactory and well-filled and are used in this study. Table 1 shows the demographic profile of the respondents.

Initial Screening and Factor Validation

Initially, the collected data set was tested to identify correctly, duplicated and poorly filled data. The poorly filled and duplicate data were removed, and finally, 213 data entries were used for the analysis. Here, we have received a 50.7% response rate, so there is less chance of non-response bias. Additionally, we conducted an examination of non-response bias using a sample of 100 early replies and 100 late responses. We assume the first 100 responses received are early, and the last 100 received responses are late. Upon conducting the non-response bias test, as outlined in Table 13 of Appendix D, it is seen that there are no significant differences among the components. Therefore, we argue that non-response bias will not affect the results of our study. The data were subjected to EFA using IBM SPSS v.20 software. Before application, EFA validation of

Table 1 Demographic Profile

Items		N	%age
Age	21–35	61	28.6
	36–50	116	54.4
	51–70	36	17.0
	Total	***213***	***100.00***
Gender	Male	137	64.3
	Female	76	35.7
	Total	213	100.00
Educational Qualification	Graduate	117	54.9
	Postgraduate	64	30.0
	Doctorate	32	15.1
	Total	***213***	***100.00***
Years of Experience	0–5	48	22.5
	5–10	76	35.6
	10–15	61	28.6
	15–20	28	13.3
	Total	***213***	***100.00***

Bolditalics signify Sum of the attributes

data is checked using the Cronbach alpha reliability test (Raut et al., 2021). The reliability test matrix, Cronbach alpha, is used to check how much the data set is reliable for the application. After the reliability check, we applied EFA on SPSS v.20, with principal component analysis, Promax rotation, and factor score limit set to not less than 0.4. From the EFA analysis, five factors were obtained from 29 items. We check the suitability of the factors and relevant items.

Hypotheses Testing and Model Validation

Confirmatory factor analysis (CFA) is utilized to measure validity and relationships across components (Nyal et al., 2023a). A conceptual model is framed with seven hypotheses from the five factors obtained using EFA. Correlation among the five factors was obtained using AMOS software v.20. To obtain correlation, factors were kept freely correlated with each other on AMOS v.20 (Kumar et al., 2023; Raut et al., 2021). After obtaining correlation among each factor, we check various validity tests such as convergent, average variance extracted (AVE), and fitness of model using other goodness and badness fit measures such as 'goodness-of-fit index (GFI),' 'root mean square error of approximation (RMSEA),' 'comparative fit index (CFI),' etc. The constructs that have been accepted, the patterns of observable variables, and the latent constructs are then utilized in the context of SEM (Nayal et al., 2021).

SEM was originally utilized in marketing theory (Bagozzi, 1980). However, its use as a theoretical framework assessment method has seen rapid growth in the social sciences and behavioral studies (MacCallum & Austin, 2000). Within a model, dependent and independent variables may be analyzed for their correlations using structural equation modeling (SEM). The suggested model and its hypotheses were tested using SEM, followed by the study of Nayal et al., (2023a, 2023b). Furthermore, it aids in systematically and thoroughly investigating a specific collection of research inquiries by concurrently analyzing the cause-and-effect connection among different constructs (Gefen et al., 2000; Gerbing & Anderson, 1988). SEM utilizes a combination of dependent variables, latent constructs, and independent factors in order to examine and evaluate multivariate data. SEM is employed to validate the statistical significance of the suggested hypotheses. The conceptual model utilized in this study is constructed within the AMOS v.20 software, and it validates the importance of the proposed hypotheses.

Data Analysis and Results

Preliminary Data Analysis

This study developed a conceptual framework for operational resilience strategy through AI. A questionnaire survey data of twenty-nine items were collected from 213 valid respondents for these five factors. The five constructs and their relevant items are shown below in Table 6 of Appendix B. The obtained survey data were first analyzed by calculating mean and variance. We did an initial analysis of the multivariate statistical test data normality. Because all two-tailed asymptotic significance is less than 0.005, the data are not normally distributed, as revealed by the one-sample K-S (Kolmogorov–Smirnov) test, shown in Table 9 of Appendix D. The survey data reliability was calculated by obtaining Cronbach alpha and was found to be 0.927. This is above the 0.7 threshold value.

EFA

The data collected from 213 respondents are used for EFA. Before approaching EFA, the first reliability of survey data was checked using Cronbach's alpha. The Bartlett's and Kaiser–Meyer–Olkin (KMO) tests were performed to test the suitability of data and was observed to be 0.908 with a $p < 0.0001$ significance level, better than the 0.6 threshold value of KMO and Bartlett's test significance of $p < 0.005$. We ensured the authenticity of responders during the data-gathering process, so there is no right or wrong answer. We used 'Harman's single-factor analysis', revealing that a single-factor represents 35.41 percent of the overall variation. There is no chance of common method biases (CMB) because this is just under 50%. As the correlation coefficient shown in Table 11 of Appendix D is less than 0.9, it satisfies that there was no issue with CMB. The EFA analysis is carried out with 'maximum likelihood' extraction, and Promax rotation is shown in Table 2. All the factors loading is above the minimum threshold value of 0.5, as suggested by (Han et al., 2021).

CFA

The CFA was performed on AMOS v.20 on five finalized constructs that are allowed to correlate with each other freely. Product development strategy and operational resilience have seven items; while, production strategy has six, artificial intelligence capability has five, and service quality strategy has four items. The obtained chi-square by the degree of freedom for the model was 1.844, which is less than the threshold limit of three used by (Raut et al., 2021). The Cronbach alpha, CR, and AVE are obtained and depicted in Table 10 of Appendix D. The obtained AVE in this study is greater than the threshold value of 0.5 suggested by Han et al. (2021) and Kumar et al. (2024). The factor loading value of measurement item PDS5 is a maximum of 0.912; while, PRS5 has the lowest loadings of 0.597. The PDS has the highest Cronbach alpha α—0.925, CR—0.925, and AVE- 0.641; while, SQS has the lowest value of Cronbach alpha α- 0.825, CR- 0.826, and AVE- 0.543. The square root of AVE shown in Table 11 of Appendix D is greater than the correlation coefficient. Finally, we employed the Heterotrait-Monotrait Ratio (HTMT) criteria, which is newly introduced. Table 12 in Appendix D reveals that all HTMT ratios are less than 0.90 (Henseler et al., 2015). As a result, the model fits the data well.

For the measurement model, CFI- 0.919, which is (CFI > 0.9) and RMSEA- 0.063, which is more than the threshold limit of 0.05, but RMSEA value between 0.05 and 0.10 is acceptable (Raut et al., 2021). Even while the GFI-0.818 is not higher than the threshold value of 0.95 for best fit, it is still within the allowed range. The standard estimate, standard error, critical ratio, and significance level obtained from CFA analysis are shown in Table 7 in Appendix C. All the items are positively correlated with their constructs and significant at less than the 0.0001 level of significance. The critical ratio is greater than 7. The standardized estimate value for all the items concerning constructs is above the threshold value of 0.5.

SEM

After validating items and constructs in the CFA analysis, SEM tests the proposed hypothesis on AMOS v.20 software. The structural model is generated on AMOS v.20 by connecting paths based on the hypothesis. The 'Chi-square/degree of freedom' for SEM is 1.847, better than the threshold of 3 for a good fit model. The model's CFI value is 0.919; while, RMSEA is 0.063, which is not less than 0.05 but between 0.05–0.1 for the best-fit model. The significance level p-value, CR value, and estimates for the paths obtained from SEM analysis are given in Table 3. The SEM with path estimates and level of significance are shown in Fig. 3. The hypothesis, *Successful production strategy has a positive relationship with firm's AI capability,'* has a maximum value of path estimate $\beta = 0.575$, significant at $p < 0.001$, and CR Value is 7.016. The hypothesis, *'Production strategy has positive linkage with successful service quality strategy,'* has the lowest path estimate value $\beta = 0.065$, with CR value 0.567. The hypotheses H1, H2, H4, H5, and H6 are significant at a p-value less than 0.001; while, H5 has $p = 0.006$ and H3 has $p = 0.057$.

Mediation Analysis

Mediation analysis is used to determine the construct's indirect influence. Partial, full, and no mediation are the three types of mediation. When the indirect influence completely overcomes the direct relation, the mediation effect occurs. Meanwhile, there is no mediation effect if the direct effect fully dominates such that the indirect effect is insignificant. When direct and indirect effects are significant, such an effect is called partial mediation. This study has one partial mediation and three full mediations (for

Table 2 Constructs, measurement items, and their loadings

Construct	Measurement Items	Loading	α	CR	AVE
Product development strategy (PDS)	PDS4	0.912	0.925	0.925	0.641
	PDS3	0.896			
	PDS5	0.892			
	PDS6	0.824			
	PDS7	0.770			
	PDS2	0.748			
	PDS1	0.653			
Production strategy (PRS)	PRS2	0.872	0.875	0.887	0.571
	PRS4	0.853			
	PRS1	0.831			
	PRS3	0.773			
	PRS6	0.651			
	PRS5	0.597			
Service quality strategy (SQS)	SQS3	0.830	0.825	0.826	0.543
	SQS1	0.814			
	SQS2	0.798			
	SQS4	0.770			
Artificial intelligence capability (AIC)	AIC2	0.876	0.847	0.869	0.574
	AIC1	0.849			
	AIC3	0.773			
	AIC4	0.699			
	AIC5	0.620			
Operational resilience (OPR)	OPR6	0.876	0.905	0.907	0.584
	OPR5	0.872			
	OPR2	0.857			
	OPR3	0.815			
	OPR7	0.737			
	OPR4	0.718			
	OPR1	0.647			

α = Cronbach alpha value; CR = Composite reliability; AVE = Average variance extracted

Table 3 Path estimates and *p*-value

Hypothesis	Standardize estimate	C.R. value	*p*-values	Accept/reject
H1	0.524	5.961	***	"Accept"
H2	0.438	3.407	***	"Accept"
H3	0.065	0.567	0.057	"Accept"
H4	0.284	3.723	***	"Accept"
H5	0.575	7.016	***	"Accept"
H6	0.139	2.769	0.006	"Accept"
H7	0.449	5.814	***	"Accept"

$p > 0.05$). Production strategy has a partial mediating effect on product development strategy and AI. AI capability fully mediates operational resilience through product development, service quality, and production strategies. The significance value and path estimate are shown in Table 4.

Fig. 3 SEM path diagram

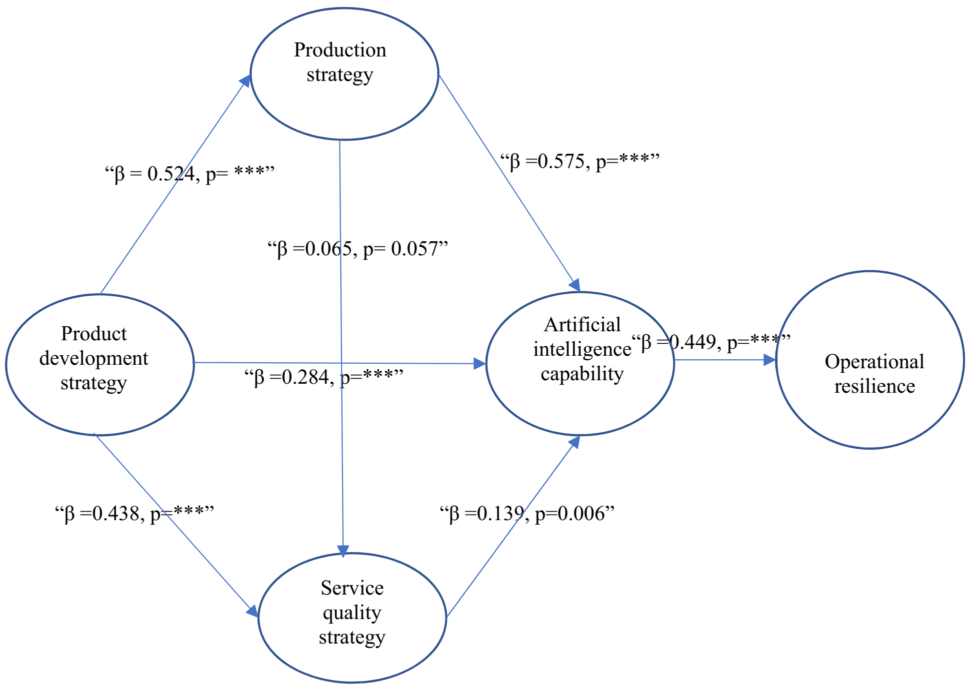


Table 4 Mediation analysis

Path	Indirect effect	Direct effect	Types of effect	Decision
PDS-PRS-AIC	0.328(0.000)	0.248(0.002)	Partial mediation	Accept
PDS-AIC-OPR	0.271(0.001)	0.123(0.098)	Full mediation	Accept
SQS-AIC-OPR	0.043(0.035)	0.139(0.097)	Full mediation	Accept
PRS-AIC-OPR	0.144(0.026)	0.257(0.054)	Full mediation	Accept

Discussion and Implications

Discussion

According to the findings, product development strategy positively impacts manufacturing strategy. However, some studies have discussed the relationship between innovation and cost or flexibility strategies, together or separately (Kumar et al., 2020; Miller & Roth., 1994; Alegre-Vidal et al., 2004) and relation of SC innovation with SC resilience in the healthcare sector (Bag et al., 2021). There may be various reasons for justifying this positive effect. The analysis of the competitive situation in the market, market size, and growth potential, the need to produce new and eco-friendly products that are first in the market, and the increasing number and type of products can support the need for improvement in firm’s ability to provide a wide range of products lines. The participation of suppliers and consumers in product development strategy emphasizes the need to increase the ability to provide a comprehensive product line depending on the current needs of consumers and the material availability. Improvement in internal and external information sharing among stakeholders increases knowledge sharing, integration, and resource sharing, thus increasing efficiency and productivity. This can help further reduce manufacturing costs and lead time and increase capacity utilization.

The implementation of a product development approach has a positive effect on the service quality strategy. The current body of research has yet to establish a conclusive correlation between product development and service quality strategies. However, prior scholarly works have indirectly examined the correlation between product development and service quality strategy by investigating the constituent elements of these constructs (Alegre-Vidal et al., 2004; Kumar et al., 2020; Sousa & da Silveira, 2020). The study and awareness of the competitive situation in the market, market size and potential of market growth, and supplier and consumer participation can improve delivery, service, reliability, and quality of products delivered to consumers. Producing new and eco-friendly products that are first in the market and increasing the number and type of products can improve competitive advantages and thus avail finance to the manufacturing firms to support after-sales service or consumer support, quality of delivery, and products. Senior management support and information sharing help improve the effectiveness of the production process, productivity, and workforce efficiency, thus further improving product quality, delivery, and reliability.

Production strategy positively influences service quality strategy. In the literature, authors have discussed the effect of delivery or quality on flexibility (Bortolotti et al., 2015). However, the literature is lacking in exploring the effect of production strategy on service quality strategy. Reduction of manufacturing lead time, setup change over time, and introduction of a comprehensive product line can improve productivity and capacity utilization. Increasing the production of products and utilizing manufacturing capabilities by a firm at a given time means operating manufacturing firms' operations at maximum efficiency. The increased efficiency of manufacturing operations means improved knowledge sharing, labor productivity, fault prediction, delivery speed, reliability, and product quality. Reducing manufacturing costs and increasing efficiency can increase financial support and develop managers' interest in investing in after-sale service and consumer support.

Successful production strategy has a significant relationship with AI capability. A successful production strategy helps firms to arrange the resources required to build AI capability. Deep neural networks can be useful for decreasing incorrect selection of design parameters, thus reducing extra modification and product development costs (Wang et al., 2021). Technologies (such as transfer learning, augmented reality, virtual reality, and virtual prototypes) can address data gaps in trial production, resulting in reduced trial production costs and improved product design quality (Wang et al., 2021). To cut waste and boost resource efficiency, an intelligent planning algorithm is highly advised for resolving low production flexibility difficulties (Chen et al., 2017; Wang et al., 2021). Production planning guidelines and heuristic scheduling algorithms effectively address inexperience with manual production scheduling (Liu et al., 2019), boosting the outputs. Human–robot collaboration can result in a wider variety and improve flexibility in the manufacturing system (Wang et al., 2021). AIC can support manufacturing execution systems using expert systems, heuristic algorithms, and processing rules, intelligent interfaces-based interactions among sensors and controllers, etc. (Huang et al., 2008). Wang et al. (2021) suggest that these methods and technologies can improve decision-making ability, increase shop-floor transparency, maximize allocation of production orders, simplify materials, and boost equipment collaboration. Manufacturers can conduct real-time monitoring of shop floor operations by utilizing a data-aware system. According to Li et al. (2018), the utilization of core algorithms, such as weighted centroid algorithms, enables this system to attain automation and provide data quality control. Additionally, it enhances the efficiency and accuracy of data collecting for shop floor information. These benefits of AI techniques and tools can reduce manufacturing cost, manufacturing lead time, and setup change over time, thus improving capacity utilization and productivity. The predictive capability and data integration (Baryannis et al., 2019; Wang et al., 2021) in manufacturing AI systems can increase the ability to provide a comprehensive product line. The execution of a successful production strategy and improvement in operational resilience are mediated by AI capability. A successful production strategy helps the industry plan accordingly and prepare for the required resources to build AIC. Using AI tools and technology in manufacturing real-time data monitoring and inventory management can improve resilience in manufacturing operations.

Our study also argues that successful product development strategy and AIC have a significant positive relationship. AIC helps with product development by studying consumer behavior using PLM data mining and AI technologies. Designers may use trend data to guide decisions that will result in a more efficient and personalized product design process. Data mining and other machine learning techniques may be used to create market risk assessment models that classify and forecast anomalous customer behavior, paving the way for introducing more environmentally friendly products. Appropriate statistical techniques for data mining have great potential to identify unmet product demands and forecast market prospects, assisting firms in modifying their product design strategies. Because augmented reality facilitates customer-generated scenarios that allow customers to produce, change, and explain their wants more easily, it can be useful for customized, collaborative design. This can lead to improved personalized designs (Wang et al., 2021). AIC may greatly improve the predictive skills necessary for demand forecasting in hazy corporate scenarios. AI-driven bots can increase customer engagement efficiency through more tailored interactions. Information integration capability (Borges et al., 2021; Nayal et al., 2021; Wang et al., 2021) enhances the role of senior management, supplier, and consumers in the product development process. AIC plays a mediation role between the successful product development strategy and manufacturing operational resilience. The product development strategy includes various design aspects associated with a large amount of data. Product development strategy is supported by data analytics resources provided by AIC, helping to improve the operational resilience of firms.

The findings indicate that AIC and service quality strategy are positively related. For efficient and desired quality and functionality, product layout design decision intelligence can be used (Wang et al., 2021). The intelligent computer-aided design (CAD) system integrated with expert systems can provide parameter recommendations to designers. Deep learning can help form an intelligent Bill of materials, affecting product quality and performance (Wang et al., 2021) Image-based inspection and analysis may be carried out using deep learning algorithms that are based on evaluation criteria incorporated with machine vision systems. Compared to manual detection, this can result in improved performance and reliability of product quality evaluation (Wang et al., 2021). Cost-effectiveness, service efficiency, and timely service may be improved using these methods, which consist of feature extraction and classification of information from consumer data and answering customer queries. Integrating distributed computing and intelligent algorithms can support real-time product monitoring. AI-based techniques, either rule-based expert systems or self-learning models, help analyze possible correlations of product failures (Wen et al., 2017), thus improving delivery reliability and reducing the number of consumer backorders. AIC has a mediation effect between service quality strategy and operational resilience.

AIC positively affects operational resilience, as it can help develop disruption absorption capacity and recoverability (Modgil et al., 2021a). Absorption of disruptions is the degree to which operations can continue as usual in the face of disruptions, without requiring major adjustments to the parts and framework of the operational process. On the other hand, recoverability ensures that the output rates of operations are restored to their usual levels following an interruption. Both of these abilities are crucial in effectively managing disruptive events (Davenport & Ronanki, 2018). According to Baryannis et al. (2019), the use of AI in organizations enables the utilization of an intelligent decision system, which aids decision-makers in making improved and expedited judgments during periods of disruption. AI-enabled collaboration, knowledge and information sharing, integration, transparency, visibility, agile decision-making, equipment, and product failure prediction, last-mile delivery, on-time intact delivery, predictive capability for demand forecasting, and risk assessment model can help manufacturing firms in developing resilience (Wang et al., 2021).

Implications

Theoretical Implications

This study contributes empirical evidence to the assertions made by Wang et al. (2021) and Fan et al. (2016) on the significance of AI capacity and SC risk information processing capability in the context of industrial operations. Our research backs the current literature in five important ways. First, the interrelations between PDS, PRS, and SQS are new in our study, as these interrelations were not empirically and directly explored in the existing literature. Researchers can use these interrelations to better understand the impact of manufacturing strategies on one another. Second, production, product development, and service quality strategies help build the manufacturing firm's AI capability. Previous literature has not studied these relationships in previous literature. The examination of these relationships will provide researchers with insights into the influence of AI on the enhancement of current production, product development, and service quality strategies. Third, the impact of building AI capability on operational resilience has not been empirically explored by authors in the literature. The insights into this effect can help researchers develop new models for AI's role in risk management. Fourth, AI capabilities are a novel mediator between the strategy for producing high-quality goods and services and the resulting operational resilience. The findings on the mediating role can assist researchers in understanding role of technology in enhancing production, PDS, and SQS; additionally, it sheds light on how this enhancement can further impact operational resilience. Fifth, this study is the first to combine AI capability with manufacturing and production strategies and their effect on operational resilience. A unique multidisciplinary model can help researchers develop integrated models on technology in combination with manufacturing and supply chain, thus addressing the need to study the connection between technology and SC resilience in manufacturing and production in today's dynamic world (Wieland & Durach, 2021).

Practical Implications

This study provides some suggestions for managers in achieving operational resilience by improving manufacturing and production strategies through building AI capability. First, managers need to focus on how product development, production, and service quality strategies support one another, as these three strategies positively impact each other. Second, production, product development, and service quality strategy support building AI capabilities in the firm using AI-based digital tools and techniques such as data mining, data-aware systems, expert systems, bots, sensors, and controllers. These AI tools and techniques improve market analysis, predictive maintenance and capability, information integration and transparency, product design, product and service quality, and reliability (Wang et al., 2021). Thus, senior managers need to promote these tools and techniques to improve all three manufacturing strategies. Third, our study argues that managers can improve a firm's operational resilience as a competitive advantage by building AI capability to help with logistics planning, real-time monitoring, and integration. However, our study goes into more depth, concluding that managers in the manufacturing industry need to focus on AI-enabled production, product development, and service quality strategy to improve operational resilience, as AI plays a mediating role in between manufacturing strategies and OPR. Hence, robust and clear-cut strategies are required to build AI capability in the manufacturing firm to gain operational resilience as a competitive advantage.

Conclusions, Limitations, and Future Research Directions

This research contributes to understanding the interrelationships between manufacturing strategies and operational resilience in the mediation of AI capability. Using a structural equation modeling approach, the interrelationship was revealed using five constructs and 29 items. Responses from 213 respondents were gathered as part of the data collection procedure. Seven hypotheses were developed and all are supported. This unique research shows how AIC can increase operational resilience by enabling interconnected product development, production, and service quality strategies. The study's findings suggest that AI has a good relationship with product development, production, and service quality strategies. Furthermore, AI-enabled techniques enhance a manufacturing organization's operational resilience. Theoretical and practical consequences have been discussed. The strongest correlation between production strategy and AIC is found in the study.

There are several avenues for additional research into operational resilience. This study focuses on three strategies, production, product development, and service quality, to explore their influence on operational resilience. In the future, new strategies may be included. Second, besides AIC, the impact of other Industry 4.0 technologies' (e.g., blockchain, internet-of-things, big data, etc.) capability for manufacturing and production strategies research may need to be investigated. Even though this study has a well-developed investigation to investigate the influence of AI-enabled techniques on operational resilience, there are some drawbacks. The objects covered in this research are not limited to the ones chosen, and some may be replaced. It is possible that the response, which came from only 213 people, may grow and provide more unique insights.

Appendix A

See Table 5.

Table 5 Available literature relevant to our study

Citation	Objective	Methodology	Country and Sector	Relevant Variables	Theory	Relevant Findings
Khan, et al. (2023)	"To study how SC data analytics enable the efficient implementation of agility, adaptability, and alignment, which are the 3As of SC risk management strategies, and thus further improve COVID-19 disruption performance."	PLS-SEM	Textile firms, Pakistan	"SC data analytics, agility, alignment, adaptability, post-COVID disruption performance"	"Dynamic capability view (DCV)"	Agility and flexibility are enhanced with SC data analytics. Furthermore, the three main components of SC risk management techniques have an impact on the interaction between SC data analytics and post-COVID-19 disruption performance
Zhao et al. (2023)	Investigate how SC digitalization impacts SC performance, with a focus on SC resilience	SEM	Manufacturing firms, China	"SC digitalization, SC resilience (absorptive capability, response capability, recovery capability)"	"Dynamic capability view (DCV)"	All components of SC resilience are favorably and profoundly impacted by SC digitization

Table 5 continued

Citation	Objective	Methodology	Country and Sector	Relevant Variables	Theory	Relevant Findings
Queiroz, et al. (2022)	To investigate the factors that contribute to SC resilience with SC readiness as a focal point	PLS-SEM	UK supply chain decision-makers from various sectors	"SC disruption orientation, SC efficiency, SC alertness, resource reconfiguration, SC resilience"	"Resource orchestration theory (ROT)"	SC awareness benefits from SC disruption orientation. Resource reconfiguration, SC efficiency, and SC resilience are significantly impacted by SC attentiveness. Rearranging resources partially moderates the link between SC resilience and attentiveness
Modgil et al. (2022)	"To study how AI implementation enhances SC resilience and how firms consider the AI scope to improve resilience."	Semi-structured interview	E-commerce supply chain, Country (Not mentioned)	Not applicable	"Not applicable"	AI can increase SC resilience by improving transparency, guaranteeing last-mile delivery, offering customized solutions to all SC participants, lessening the effect of disruption, and supporting an agile procurement strategy
Munir et al. (2022)	"To identify and explore the role of essential capabilities to address disruptions like COVID-19 to enable SC resilience and responsiveness to improve performance further."	SEM	Manufacturing firms such as textile, apparel, pharmaceutical, chemicals, food processing and FMCGs, Pakistan	"Data analytic capability, improvisation capability, anticipation capability, SC resilience"	"Information processing theory (IPT)"	Data analysis skills positively impact the capacity for foresight and improvisation, and these capacities further enhance supply chain resilience. The capacity for foresight and improvisation is a mediator in the link between SC resilience and data analytic competence
Khan et al. (2022)	To investigate the impact of COVID-19 on sustainable construction performance, smart technology, green practices, and crisis mitigation solutions for the SC	SEM	Manufacturing firms, Pakistan	"COVID-19 pandemic, SC crisis mitigation strategies, smart technologies"	None	Smart technology and SC crisis mitigation techniques are associated with a decrease in the spread of the COVID-19 pandemic
Baabdullah et al. (2021)	To empirically explore the causes and effects of business-to-business (B2B) SMEs' effective adoption of AI methods	SEM	B2B SMEs in Saudi Arabia	"SME's AI practices, customer engagement, and service experience"	"Technology organization-environment (TOE) model."	With AI and AI-based customer involvement, B2B SMEs' AI practices positively support service experience
Nayal et al. (2021)	To investigate the factors driving AI adoption and the ways in which AI impacts supply chain risk mitigation (SCRM)	SEM	Agriculture SC firms in India	"AI, SC risk mitigation"	"Technology, organization, and environment (TOE) framework"	"AI positively impacts SC risk mitigation."

Table 5 continued

Citation	Objective	Methodology	Country and Sector	Relevant Variables	Theory	Relevant Findings
Belhadi et al. (2021b)	Examine how supply chain volatility and uncertainty affect AI, SC resilience, and SC performance from both upstream and downstream perspectives	SEM	Digitalized firms from multi-sector in North Africa, South Europe, and Southern Asia	"Supply chain resilience, AI"	"Organizational information processing theory (OIPT)"	"AI affects SC collaboration and SC collaboration further impacts SC resilience or SC collaboration mediate the relationship between AI and SC resilience."
Avella et al. (2011)	To further show the cumulative nature of the four key manufacturing goals—quality, delivery, flexibility, and cost-efficiency by examining the theoretical assumptions and empirical data relevant to the sand cone and trade-off models	SEM	Manufacturers in Spain	"Quality, delivery, flexibility, cost-efficiency"	"Sand cone model, competitive progression theory"	Through delivery, quality has a positive indirect effect on adaptability; similarly, adaptability and environmental preservation all have positive indirect effects on cost-effectiveness
Li et al. (2005)	To investigate how management control systems influence decisions about product development and processes and how environmental complexity influences management control system selection	SEM	Firms in China	"Product development, manufacturing process"	None	Product development improves manufacturing

Appendix B

See Table 6.

Table 6 List of measures of variables in the conceptual model

Variables	Items	Authors
Production strategy (PRS)	Low manufacturing cost	(Kumar et al., 2020)
	Manufacturing products with high productivity	(Kumar et al., 2020)
	Having the capability to improve capacity utilization	(Kumar et al., 2020)
	Reduce setup changeover time	(Ehie, Muogboh. 2016)
	Having the ability to provide a comprehensive product line	(Kumar et al., 2020)
	Reduce manufacturing lead time	(Ehie, Muogboh. 2016)
Product development strategy (PDS)	The competitive situation in the market	(Morita et al.,. 2018)
	Market size and growth potential	(Morita et al.,. 2018)
	Senior management support	(Morita et al.,. 2018)
	Suppliers and customers participation	(Morita et al.,. 2018)
	Internal and external information sharing, including cross-functional interactions	(Morita et al.,. 2018)
	Producing new eco-friendly products that are first in the market	(Helo & Hao., 2022)
	Increasing number and type of products or services	(Helo & Hao., 2022)

Table 6 continued

Variables	Items	Authors
Service quality strategy (SQS)	Improving product reliability	(Ehie, Muogboh. 2016)
	Improve after-sales service or consumer support	(Ehie, Muogboh. 2016)
	Increase delivery reliability	(Ehie, Muogboh. 2016)
	The frequency of customer backorders is low	(Kaur et al., 2017)
Artificial intelligence capability (AIC)	Technological infrastructure	(Chowdhury et al., 2022)
	Data resources	(Chowdhury et al., 2022)
	AI skills	(Chowdhury et al., 2022)
	Man–machine cooperation	(Wang et al., 2021)
	Intelligent logistics planning	(Wang et al., 2021)
Operational resilience (OPR)	Restoring to normal operations is not time-consuming	(Essuman et al., 2020)
	Quickly recovers to its regular operations	(Essuman et al., 2020)
	Effectively restores operations to normal	(Essuman et al., 2020)
	The firm can carry out its functions despite some damage done to it	(Essuman et al., 2020)
	Without much deviation, we can meet average operational and market needs	(Essuman et al., 2020)
	The firm gives us much time to consider a reasonable response	(Essuman et al., 2020)
	Our organization operates effectively in many instances without adjustments	(Essuman et al., 2020)

Appendix C

See Table 7, 8.

Table 7 CFA path estimate

			Unstandardized Estimate	Standardized estimate	S.E	C.R	P
AIC1	<–	AIC	1.000	0.791			
AIC2	<–	AIC	.953	0.763	0.081	11.699	***
AIC3	<–	AIC	1.022	0.816	0.081	12.691	***
AIC4	<–	AIC	1.075	0.804	0.086	12.464	***
AIC5	<–	AIC	.834	0.592	0.096	8.704	***
OPR1	<–	OPR	1.000	0.583			
OPR2	<–	OPR	1.602	0.851	0.176	9.089	***
OPR3	<–	OPR	1.377	0.753	0.163	8.445	***
OPR4	<–	OPR	1.408	0.730	0.170	8.272	***
OPR5	<–	OPR	1.632	0.827	0.183	8.941	***
OPR6	<–	OPR	1.672	0.837	0.186	9.007	***
PRS5	<–	PRS	1.000	0.706			
PRS4	<–	PRS	1.095	0.808	0.099	11.056	***
PRS3	<–	PRS	1.092	0.782	0.102	10.710	***
PRS2	<–	PRS	1.091	0.844	0.095	11.518	***
PRS1	<–	PRS	1.016	0.829	0.090	11.326	***
SQS1	<–	SQS	1.000	0.710			
SQS2	<–	SQS	1.042	0.763	0.110	9.487	***
SQS3	<–	SQS	1.022	0.759	0.108	9.453	***
SQS4	<–	SQS	1.019	0.713	0.113	9.009	***
PDS1	<–	PDS	1.000	0.718			
PDS2	<–	PDS	1.175	0.860	0.095	12.388	***

Table 7 continued

			Unstandardized Estimate	Standardized estimate	S.E	C.R	P
PDS3	<–	PDS	1.240	0.893	0.096	12.879	***
PDS4	<–	PDS	1.277	0.920	0.096	13.264	***
PDS5	<–	PDS	1.185	0.829	0.099	11.944	***
PDS6	<–	PDS	1.036	0.707	0.102	10.144	***
PDS7	<–	PDS	.866	0.635	0.095	9.078	***
PRS6	<–	PRS	.859	0.515	0.121	7.122	***
OPR7	<–	OPR	1.325	0.736	0.159	8.316	***

Table 8 SEM path estimate

			Unstandardized estimate	Standardized estimate	S.E	C.R	P
PRS	<–	PDS	0.524	0.483	0.088	5.961	***
SQS	<–	PDS	0.438	0.312	0.129	3.407	***
SQS	<–	PRS	0.065	0.050	0.115	0.567	0.057
AIC	<–	PDS	0.284	0.255	0.076	3.723	***
AIC	<–	PRS	0.575	0.559	0.082	7.016	***
AIC	<–	SQS	0.139	0.174	0.050	2.769	0.006
OPR	<–	AIC	0.449	0.512	0.077	5.814	***
AIC1	<–	AIC	1.000	0.788			
AIC2	<–	AIC	0.955	0.761	0.082	11.660	***
AIC3	<–	AIC	1.023	0.814	0.081	12.633	***
AIC4	<–	AIC	1.076	0.802	0.087	12.411	***
AIC5	<–	AIC	0.835	0.591	0.096	8.677	***
OPR1	<–	OPR	1.000	0.583			
OPR2	<–	OPR	1.603	0.852	0.176	9.105	***
OPR3	<–	OPR	1.378	0.754	0.163	8.459	***
OPR4	<–	OPR	1.407	0.730	0.170	8.283	***
OPR5	<–	OPR	1.627	0.825	0.182	8.938	***
OPR6	<–	OPR	1.669	0.837	0.185	9.012	***
PRS5	<–	PRS	1.000	0.705			
PRS4	<–	PRS	1.096	0.807	0.100	11.015	***
PRS3	<–	PRS	1.093	0.781	0.102	10.674	***
PRS2	<–	PRS	1.095	0.846	0.095	11.500	***
PRS1	<–	PRS	1.021	0.831	0.090	11.322	***
SQS1	<–	SQS	1.000	0.712			
SQS2	<–	SQS	1.034	0.759	0.109	9.472	***
SQS3	<–	SQS	1.020	0.760	0.108	9.476	***
SQS4	<–	SQS	1.017	0.714	0.113	9.030	***
PDS1	<–	PDS	1.000	0.718			
PDS2	<–	PDS	1.174	0.859	0.095	12.385	***
PDS3	<–	PDS	1.239	0.892	0.096	12.877	***
PDS4	<–	PDS	1.277	0.920	0.096	13.273	***
PDS5	<–	PDS	1.185	0.830	0.099	11.950	***
PDS6	<–	PDS	1.036	0.708	0.102	10.152	***
PDS7	<–	PDS	0.866	0.635	0.095	9.086	***
PRS6	<–	PRS	0.857	0.512	0.121	7.077	***
OPR7	<–	OPR	1.322	0.735	0.159	8.315	***

Appendix D

See Table 9, 10, 11, 12, 13.

Table 9 One-sample Kolmogorov–Smirnov test for normality of distribution

	N	Normal parameters[a,b]		Most extreme differences			Kolmogorov–Smirnov Z	Asymp. Sig. (2-tailed)
		Mean	Std. deviation	Absolute	Positive	Negative		
SQS1	213	2.6056	0.96366	0.275	0.275	− 0.171	4.014	0.000
SQS2	213	3.1127	0.93490	0.270	0.169	− 0.270	3.941	0.000
SQS3	213	2.4225	0.92137	0.278	0.278	− 0.187	4.053	0.000
SQS4	213	2.6103	0.97772	0.288	0.288	− 0.182	4.200	0.000
OPR1	213	2.6948	0.82177	0.275	0.275	− 0.175	4.017	0.000
OPR2	213	2.8122	0.90201	0.253	0.253	− 0.160	3.688	0.000
OPR3	213	2.6479	0.87575	0.287	0.287	− 0.187	4.185	0.000
OPR4	213	3.0047	0.92399	0.207	0.195	− 0.207	3.017	0.000
OPR5	213	2.9531	0.94552	0.224	0.224	− 0.195	3.263	0.000
OPR6	213	2.9718	0.95619	0.226	0.226	− 0.206	3.292	0.000
OPR7	213	2.8310	0.86307	0.241	0.241	− 0.169	3.512	0.000
AIC1	213	3.9812	0.69338	0.384	0.325	− 0.384	5.605	0.000
AIC2	213	3.8732	0.68535	0.381	0.309	− 0.381	5.559	0.000
AIC3	213	4.0704	0.68661	0.365	0.330	− 0.365	5.331	0.000
AIC4	213	3.9296	0.73312	0.346	0.288	− 0.346	5.046	0.000
AIC5	213	3.7653	0.77174	0.314	0.244	− 0.314	4.588	0.000
PRS1	213	3.9296	0.65135	0.355	0.311	− 0.355	5.185	0.000
PRS2	213	3.9296	0.68661	0.325	0.290	− 0.325	4.742	0.000
PRS3	213	3.7934	0.74261	0.356	0.278	− 0.356	5.196	0.000
PRS4	213	3.7840	0.72054	0.341	0.270	− 0.341	4.974	0.000
PRS5	213	3.7840	0.75256	0.312	0.251	− 0.312	4.560	0.000
PRS6	213	3.3005	0.88693	0.221	0.196	− 0.221	3.232	0.000
PDS1	213	4.0516	0.68140	0.320	0.300	− 0.320	4.664	0.000
PDS2	213	3.9765	0.66900	0.350	0.312	− 0.350	5.103	0.000
PDS3	213	3.9906	0.67983	0.332	0.302	− 0.332	4.842	0.000
PDS4	213	3.9812	0.67964	0.323	0.296	− 0.323	4.717	0.000
PDS5	213	4.0376	0.69939	0.295	0.282	− 0.295	4.313	0.000
PDS6	213	4.0751	0.71638	0.322	0.293	− 0.322	4.701	0.000
PDS7	213	4.1080	0.66771	0.287	0.287	− 0.271	4.192	0.000

Table 10 Convergent validity

	CR	AVE	MSV	MaxR(H)
SQS	0.826	0.543	0.130	0.828
AIC	0.869	0.574	0.506	0.881
OPR	0.907	0.584	0.238	0.919
PRS	0.887	0.571	0.506	0.905
PDS	0.925	0.641	0.329	0.945

CR composite reliability; AVE- average variance explained; MSV- maximum shared variance

Table 11 Discriminant validity

	SQS	AIC	OPR	PRS	PDS
SQS	**0.737**				
AIC	0.361	**0.758**			
OPR	0.304	0.488	**0.764**		
PRS	0.201	0.711	0.424	**0.756**	
PDS	0.337	0.574	0.392	0.483	**0.801**

Bold numbers are the square root of AVE,

Table 12 Heterotrait-Monotrait Ratio-HTMT

	SQS	OPR	AIC	PRS	PDS
SQS					
OPR	0.313				
AIC	0.356	0.504			
PRS	0.209	0.468	0.739		
PDS	0.34	0.39	0.589	0.484	

Table 13 Non-response bias test

		Mean	N	Std. Deviation	Std. Error Mean	t	Sig. (2-tailed)
Paired samples statistics							
Pair 1	SQS_E	3.1411	100	0.77113	0.07711	3.026	0.103
	SQS_L	2.8340	100	0.89259	0.08926		
Pair 2	OPR_E	3.2567	100	0.74112	0.07411	4.432	0.100
	OPR_L	2.8666	100	0.84940	0.08494		
Pair 3	AIC_E	3.7063	100	0.79410	0.07941	− .586	0.559
	AIC_L	3.7652	100	0.82002	0.08200		
Pair 4	PRS_E	3.5152	100	0.78147	0.07815	− 1.795	0.076
	PRS_L	3.6747	100	0.72902	0.07290		
Pair 5	PDS_E	3.7191	100	0.77600	0.07760	− 1.437	0.154
	PDS_L	3.8486	100	0.77142	0.07714		

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Declarations

Conflict of interest The authors declare that they have no competing interests.

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Key Questions

1. How does product development strategy influence production and service quality strategies?
2. How does production strategy influence service quality strategy?
3. How do production, product development, and service quality strategies contribute to building AI capability in the manufacturing industry?
4. How does AI capability enhance the competitiveness of manufacturing firms in terms of operational resilience?

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